

RC2-Style Full-Repo Convergence Plan

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Objective

Converge this repository toward a RollerCoaster Tycoon-style engineering posture: deterministic behavior, tight control of state transitions, performance-first loops, and low-overhead architecture while preserving correctness and safety.

Non-Negotiable Safety Rules

- ? No destructive rewrites of the full repo in one pass.
- ? No behavior regressions accepted without explicit signoff.
- ? Refactor in small validated batches with rollback points.
- ? All high-risk modules require tests before and after changes.

12-Pronged Convergence Workstream

1. Hotspot profiling and baseline performance map
2. Data layout hardening and state minimization
3. Deterministic update loop standardization
4. Branch and memory-access simplification
5. I/O boundary isolation
6. Error-path tightening and explicit fail states
7. Telemetry cost reduction
8. Concurrency and ordering discipline
9. Config and constant table normalization
10. Test harness expansion for invariants
11. Docs-to-code drift elimination
12. Release gate hardening with benchmark checks

Execution Epochs

Epoch	Scope	Output	Validation
E1	Core runtime hotspots	Refactor patchset + perf notes	Unit tests + microbench
E2	Policy/gates/telemetry	Deterministic guardrails	Regression + gate checks
E3	App surfaces and integrations	Low-overhead interface cleanup	Integration tests
E4	Repo-wide convergence	Final consistency pass	Full CI + benchmark delta

Acceptance Criteria

- ? All modified modules preserve functional behavior under existing tests.
- ? Measured performance improves or remains neutral in key paths.
- ? No increase in incident-prone complexity at system boundaries.
- ? Convergence artifacts include clear rollback and change logs.

Founder Summary

We will make the codebase feel assembly-disciplined without reckless rewrite risk: small passes, hard validation, measurable speed/control gains, and preserved reliability.